

DDoS Attack Detection Using Machine Learning and Deep Learning: An Automated Approach with SelectKBest and LSTM

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ABSTRACT—Due to the sharp increase in data volume, significant security concerns have arisen regarding Distributed Denial of Service (DDoS) attacks, which inflict severe financial and reputational losses on organizations worldwide. While extensive literature addresses DDoS detection, existing frameworks often rely on generic predictions or lack automated feature engineering pipelines. To resolve this limitation, this paper proposes an optimized computational framework that automates feature selection using the SelectKBest algorithm, integrated with a Long Short-Term Memory (LSTM) recurrent neural network for granular attack classification. The proposed methodology was trained and validated across four prominent vector categories: UDP-Lag, SYN, LDAP, and NetBIOS, extracted from the benchmark CICDDoS2019 dataset. Experimental evaluations demonstrate that the proposed architecture achieves robust performance, with accuracy and evaluation metrics consistently spanning between 98% and 99% in detecting and isolating these advanced cyber threats.

KEYWORDS—Cyber Security, DDoS Attacks, SelectKBest, LSTM, CICDDoS2019.

I. INTRODUCTION

Cyber security has emerged as an essential cornerstone of the contemporary digitalized society. It encompasses the study and implementation of technological controls, protocols, and processes designed to protect computational assets, networks, programs, and data from internal or external malicious vectors and unauthorized access [1]. In modern technological ecosystems, integrating network infrastructure defense mechanisms with artificial intelligence—specifically utilizing Machine Learning and Deep Learning—endows systems with adaptive traits. This facilitates the accurate detection of known attack signatures while uncovering behavioral anomalies that deviate from baseline network statistics [2].

With the massive expansion of global network traffic and the exponential increase in transactional data volumes, vulnerabilities related to system security and user privacy have grown rapidly. One of the most disruptive threats operating on the Internet today is the Distributed Denial of Service (DDoS) attack. These incursions carry the intrinsic potential to inflict massive financial and reputational damages on enterprise networks. Institutional reports underline the severity of this threat: industry data from

A. Machine Learning and Classification

Machine Learning grants computing systems the ability to extract implicit patterns and operational decision rules directly from historical data reservoirs [6]. Among the conventional classification algorithms mapped for intrusion detection, Support Vector Machines (SVM) and the Naive Bayes classifier stand out [7]. The conditional probability calculation of a specific class given a network feature vector is formalized by equation (1):

$$P(A|B) = [P(B|A) \cdot P(A)] / P(B) \quad (1)$$

Where $P(A|B)$ denotes the posterior probability of the event given the telemetry evidence observed within the network topology.

A. Deep Learning and LSTM Architecture

Deep Learning extends artificial neural network capabilities by stacking hidden layers of mathematical processing to automatically capture abstract representations from raw inputs [8]. Since network packet flows constitute time series with strong sequential correlations, Recurrent Neural Networks (RNN) offer high efficacy but suffer from vanishing gradient limitations over long sequences. To mitigate this constraint, a Long Short-Term Memory (LSTM) architecture is employed [9]. LSTM cells embed internal gating mechanisms (forget, input, and output gates) that work in synergy to discard irrelevant data and retain long-term structural dependencies.

I. RELATED WORK

Prior literature displays diverse approaches toward using intelligence frameworks to counter network intrusions. Alshamrani et al. [10] integrated classification models within Software-Defined Networking (SDN) environments to pull real-time telemetry. Seyda and Nazife [11] focused on processing logs from corporate firewalls integrated with SIEM platforms, applying the SMO classifier. Within deep learning domains, Yuan et al. [12] introduced the *DeepDefense* framework, demonstrating empirically that deep architectures outperform traditional linear statistical models by natively isolating temporal correlations embedded in packet headers.

Akamai reveals that approximately 40% of organizations targeted by DDoS attacks belong to the global financial services sector, while consolidated analytical data from Kaspersky Lab indicates an 84% surge in attack occurrences over equivalent historical periods. This confirms that malicious actors are systematically refining their tools to maximize the persistence and traffic volume of network offensives.

Given the wide array of DDoS attack variants and their expanding structural complexity, the engineering of high-throughput, efficient detection and mitigation frameworks has become critical [3]. Although traditional literature features established models utilizing conventional classifiers such as Naive Bayes, Support Vector Machines (SVM), and dense Artificial Neural Networks, a distinct gap remains concerning automated feature optimization paired with recurrent, time-aware topographies. This paper addresses this gap by introducing a quantitative computational framework that pairs the statistical SelectKBest algorithm for feature engineering with a Long Short-Term Memory (LSTM) recurrent neural network architecture. The primary objective of this investigation is to validate an optimized model capable of swiftly identifying four specific DDoS attack vectors (UDP-Lag, SYN, LDAP, and NetBIOS) from the public benchmark CICDDoS2019 dataset while maximizing operational metrics.

I. THEORETICAL BACKGROUND

The foundations of this research bridge telecom telemetry and deep predictive modeling.

A. Distributed Denial of Service (DDoS)

A Distributed Denial of Service (DDoS) attack is a large-scale coordinated offensive aimed at exhausting the computational resources or available bandwidth of a target system by flooding it with massive volumes of concurrent requests [4]. The central objective is to render legitimate services unavailable to authorized users by depleting server memory, buffer queues, or processing units.

DDoS attacks are classically divided into two major macro-categories: Flooding Attacks, which operate through the continuous generation of raw network packets directed at specific ports or protocols (such as SYN flooding exploiting the TCP handshake state tables); and Amplification and Reflection Attacks. The latter exploit flaws in application-layer protocols (such as LDAP or NetBIOS) to redirect asymmetric traffic from legitimate third-party servers toward the victim using spoofed source IPs [5].

I. PROPOSED METHODOLOGY

The proposed approach is built upon Knowledge Discovery in Databases (KDD) paradigms and follows three main stages:

A. Data Acquisition and Pre-processing

Raw network streams undergo an intensive data cleaning phase. This step enforces the explicit removal of socket-level identifiers (such as *Flow ID*, *Source IP*, *Source Port*, *Destination IP*, *Destination Port*, *Protocol*, and *SimilarHTTP*), alongside chronological indicators (*Timestamp*). Stripping these variables ensures classifier neutrality against geographical and temporal bias. Null values (*NaN*) are discarded and remaining features are normalized via linear scaling into a strict $[-1, 1]$ boundary.

A. Automated Feature Selection with SelectKBest

To combat the curse of dimensionality, the SelectKBest statistical algorithm is deployed. This mechanism automatically computes the mathematical relevance of each of the 80 network parameters against the target label. The evaluation metric chosen was $f_{classif}$, which computes the ANOVA F-value (Analysis of Variance) score across predictors to rank features and retain only the top k informative variables.

A. LSTM Deep Neural Network Architecture

The optimized feature vector feeds a five-layer sequential deep model: an entry layer with 20 LSTM memory cells; three intermediate recurrent layers containing 10 LSTM units each; coupled with a strict 30% (0.3) Dropout regularization rate to thwart overfitting; and a final dense output layer utilizing a *Sigmoid* activation function for precise binary classification. The loss function was parameterized as Mean Squared Error (MSE), optimized via the *RMSprop* training algorithm.

I. EXPERIMENTS AND EVALUATION

The experimental setup was deployed on the Google Colab platform utilizing GPU hardware accelerators and a Python pipeline. Validation was conducted on the benchmark international CICDDoS2019 dataset [13]. Training sessions spanned a fixed window of 50 epochs per attack class.

System performance was measured via Confusion Matrix extraction to capture Accuracy, Precision, Recall, and F1-Score. Consolidated quantitative results are summarized in Table I.

TABLE I
PERFORMANCE METRICS OF THE OPTIMIZED MODEL PER DDOS
VECTOR

DDOS ATTACK	CATEGORY	ACCURACY	F1-SCORE
Reflection	98.85%	0.988	
Reflection	99.12%	0.991	
Exploitation	98.54%	0.985	
Exploit.	99.03%	0.990	

The analysis of Table I highlights the robustness of the developed topography, yielding general accuracy bounds firmly situated between 98% and 99%. Low false-alarm counts are further verified by consistently high F1-Scores.

I. CONCLUSION

This study successfully introduced an optimized scientific framework pairing the SelectKBest statistical algorithm with deep LSTM recurrent models for automated DDoS detection. The architecture achieved excellent predictive metrics within the 98%–99% margin on the CICDDoS2019 dataset. Future investigations should evaluate this computing system within high-speed live data streams.

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